Thus we have the following allocations:

	I	II	III	IV	a_i
A	(6) 5	(6) 1	(17) 3	(5) 3	34
B	(15) 3	3	5	4	15
C	6	4	4	(12) 3	12
D	4	(19) 1	4	5	19
b_j	21	25	17	17	

Transportation schedule :

$A \rightarrow$ I, $A \rightarrow II$, $A \rightarrow III$, $A \rightarrow IV$, $B \rightarrow I$, $C \rightarrow IV$, $D \rightarrow II$

$= (6\times5)+(6+1)+(17\times3)+(5\times3)$

$(15\times3)+(12\times3)+(19\times1)$

$= 30+6+51+15+45+36+19$

= ₹ 202

Exercise 10.1

1. What is transportation problem?
2. Write mathematical form of transportation problem.
3. what is feasible solution and non degenerate solution in transportation problem?
4. What do you mean by balanced transportation problem?
5. Find an initial basic feasible solution of the following problem using north west corner rule.

	D_1	D_2	D_3	D_4	Supply
O_1	5	3	6	2	19
O_2	4	7	9	1	37
O_3	3	4	7	5	34
Demand	16	18	31	25	

6. Determine an initial basic feasible solution of the following transportation problem by north west corner method.

	Bangalore	Nasik	Bhopal	Delhi	Capacity
Chennai	6	8	8	5	30
Madurai	5	11	9	7	40
Trichy	8	9	7	13	50
Demand (Units/day	35	28	32	25	

7. Obtain an initial basic feasible solution to the following transportation problem by using least- cost method.

	D_1	D_2	D_3	Supply
O_1	9	8	5	25
O_2	6	8	4	35
O_3	7	6	9	40
demand	30	25	45	

8. Explain Vogel's approximation method by obtaining initial feasible solution of the following transportation problem.

	D_1	D_2	D_3	D_4	Supply
O_1	2	3	11	7	6
O_2	1	0	6	1	1
O_3	5	8	15	9	10
Demand	7	5	3	2	

9. Consider the following transportation problem.

	D_1	D_2	D_3	D_4	Availability
O_1	5	8	3	6	30
O_2	4	5	7	4	50
O_3	6	2	4	6	20
Requirement	30	40	20	10	

Determine initial basic feasible solution by VAM.

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